采用单位振幅单色平面波垂直照射孔径.

$$U_0(x_0, y_0) = 1$$

$$U_0'(x_0, y_0) = t(x_0, y_0)$$

由夫琅禾费衍射公式 (4.4.14), 在距衍射屏 d 处观察平面上光场复振幅为:

$$U_i(x_i, y_i) = \frac{e^{ikd}}{i\lambda d} e^{i\pi\lambda d(\xi^2+\eta^2)} F\{t(x_0, y_0)\}_{\xi=\frac{x_i}{\lambda d},\ \eta=\frac{y_i}{\lambda d}}$$

$$= \frac{e^{ikd}}{i\lambda d} e^{i\pi\lambda d(\xi^2+\eta^2)} T(\xi, \eta)$$

其中, $T(\xi, \eta)$ 为物体空间频谱函数. 即透射率的FT.

$$T(\xi, \eta) = \iint_{-\infty}^{\infty} t(x_0, y_0) e^{-i2\pi(\xi x_0 + \eta y_0)} dx_0 dy_0$$

光强:

$$I_i(x_i, y_i) = |U_i(x_i, y_i)|^2 = \frac{T^2(\xi, \eta)}{(\lambda d)^2} \propto T^2(\xi, \eta).$$

矩形孔 & 狭缝

1. 矩形孔

$$U_i(x_i,y_i) = \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}(x_i^2+y_i^2)} F\{t(x_0,y_0)\}$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}(x_i^2+y_i^2)} \iint_{\Sigma_0} t(x_0,y_0) e^{-i\frac{2\pi}{\lambda d}(x_i x_0 + y_i y_0)} dx_0 dy_0$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}(x_i^2+y_i^2)} \int_{-\frac{a}{2}}^{\frac{a}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{-\frac{i2\pi}{\lambda d}(x_i x_0 + y_i y_0)} dx_0 dy_0$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}(x_i^2+y_i^2)} ab\,\mathrm{sinc}\left(\frac{ax_i}{\lambda d}, \frac{by_i}{\lambda d}\right)$$

光强，

$$I_i(x_i,y_i) = |U_i(x_i,y_i)|^2 = \left(\frac{ab}{\lambda z}\right)^2 \mathrm{sinc}^2\left(\frac{ax_i}{\lambda z}, \frac{by_i}{\lambda z}\right)$$

$$= I_{i0}\,\mathrm{sinc}^2\left(\frac{ax_i}{\lambda z}, \frac{by_i}{\lambda z}\right)$$

其中，

$$I_{i0} = I_{i0}(0,0) = \left(\frac{ab}{\lambda z}\right)^2$$

2. 单缝夫琅禾费衍射.

相当于一维矩形孔

$$U_i(x_i) = \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}x_i^2} F\{t(x_0)\}$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ik}{2d}x_i^2} \iint_{\Sigma_0} t(x_0) e^{-\frac{i2\pi}{\lambda d}x_i x_0} dx_0$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ikd}{2d}x_i^2} \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-\frac{i2\pi}{\lambda d}x_i x_0} dx_0$$

$$= \frac{e^{ikd}}{i\lambda d} e^{\frac{ikd}{2d}x_i^2} a\,\mathrm{sinc}^2\left(\frac{ax_i}{\lambda d}\right)$$

$$I_i(x_i) = |U_i(x_i)|^2 = \left(\frac{a}{\lambda d}\right)^2 \mathrm{sinc}^2\left(\frac{ax_i}{\lambda d}\right) = I_{i0}\,\mathrm{sinc}^2\left(\frac{ax_i}{\lambda d}\right)$$

其中,

$$I_{i0} = I_{i0}(0) = \left(\frac{a}{\lambda d}\right)^2$$

衍射图样为一组方向与单缝上的线状衍射条纹.

在 x_i (y_i) 上第一个零点位置由

$$\frac{ax_i}{\lambda d} = \pm 1 \quad \left(\frac{by_i}{\lambda d} = \pm 1\right)$$

确定.

$$\Rightarrow \qquad x_i = \pm\frac{\lambda d}{a}, \quad y_i = \pm\frac{\lambda d}{b}.$$

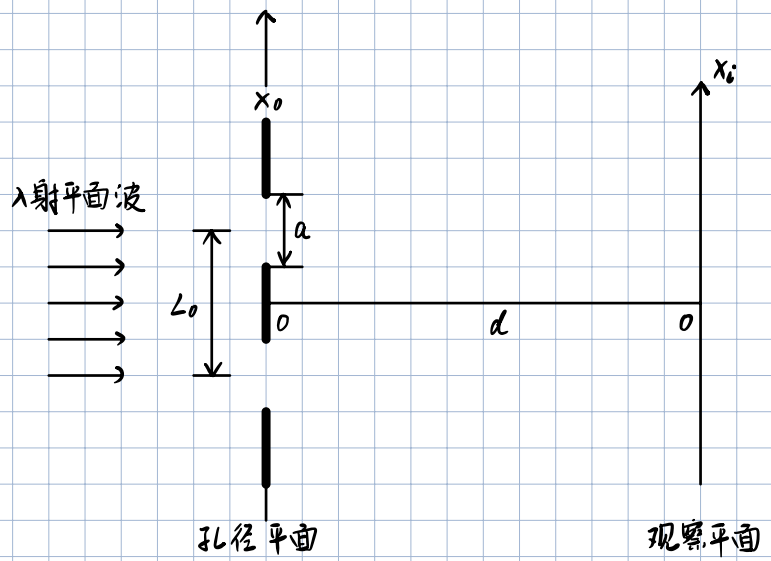
主瓣宽度:

$$\Delta x_i = \frac{2\lambda d}{a} \qquad \Delta y_i = \frac{2\lambda d}{b}$$

在中央最大值2边,周期性地出现零点,

$$T = \frac{\lambda d}{a} \quad \left(T = \frac{\lambda d}{b}\right)$$

3. 双狭缝



$$t(x_0) = rect\left(\frac{x_0 - L_0/2}{a}\right) + rect\left(\frac{x_0 + L_0/2}{a}\right)$$

$$U_i(x_i) = \frac{e^{ikd}}{\lambda d} e^{\frac{ik}{2d}x_i^2} F\{t(x_0)\}$$

$$= \frac{e^{ikd}}{\lambda d} e^{\frac{ik}{2d}x_i^2} \left[\int_{-(a+L_0)/2}^{(a+L_0)/2} e^{-i\frac{2\pi x_i x_0}{\lambda d}} dx_0 + \int_{-(a-L_0)/2}^{(a-L_0)/2} e^{-i\frac{2\pi x_i x_0}{\lambda d}} dx_0\right]$$

$$= \frac{e^{ikd}}{\lambda d} e^{\frac{ik}{2d}x_i^2} L_0 \, sinc\left(\frac{a x_i}{\lambda d}\right)\left(e^{-i\frac{\pi L_0 x_i}{\lambda d}} + e^{i\frac{\pi L_0 x_i}{\lambda d}}\right)$$

$$= \frac{e^{ikd}}{\lambda d} e^{\frac{ik}{2d}x_i^2} L_0 \, sinc\left(\frac{a x_i}{\lambda d}\right) \cos\left(\frac{\pi L_0 x_i}{\lambda d}\right)$$

光强，

$$I_i(x_i) = |U_i(x_i)|^2 = \left(\frac{L_0}{\lambda d}\right)^2 sinc^2\left(\frac{a x_i}{\lambda d}\right) \cos^2\left(\frac{\pi L_0 x_i}{\lambda d}\right)$$

双狭缝的衍射图为2个无限窄狭缝的杨氏干涉与一有限宽狭缝衍射图的乘积. 为一种干涉与衍射综合效应.

4. 多缝.

$$t(x_0)=\sum_{m=0}^{M-1}\mathrm{rect}\left(\frac{x_0-nL_0}{a}\right)=\sum_{n=1}^{\frac{N}{2}}\left\{\mathrm{rect}\left[\frac{x_0-(2n-1)\frac{L_0}{2}}{a}\right]+\mathrm{rect}\left[\frac{x_0+(2n-1)\frac{L_0}{2}}{a}\right]\right\}$$

$\frac{N}{2}$ 个双狭缝间叠加.

衍射光场:

$$U_i(x_i)=\frac{e^{ikd}}{i\lambda d}e^{\frac{ik}{2d}x_i^2}F\{t(x_0)\}$$

$$=\cdots$$

$$=\frac{e^{ikd}}{i\lambda d}e^{\frac{ik}{2d}x_i^2}\mathrm{sinc}\left(\frac{ax_i}{\lambda d}\right)\sum_{m=0}^{M}e^{-i\frac{2\pi mL_0x_i}{\lambda d}}$$

等比级数

$$\sum_{m=0}^{M}e^{-i2\pi mL_0\xi}=\frac{1-e^{-i2\pi\xi ML_0}}{1-e^{-i2\pi\xi_0L_0}}=e^{-i(M-1)\pi\xi}\frac{\sin(\pi ML_0\xi)}{\sin(\pi\xi L_0)}\qquad\left(\xi=\frac{x_i}{\lambda d}\right)$$

$\therefore$

$$U_i=\frac{L_0e^{ikd}e^{i\pi\lambda\xi^2}e^{-i(M-1)\pi\xi L_0}}{i\lambda d}\mathrm{sinc}(a\xi)\frac{\sin(\pi ML_0\xi)}{\sin(\pi L_0\xi)}$$

衍射光强:

$$I_i(\xi)=|U_i(\xi)|^2=\left(\frac{a}{\lambda d}\right)^2\mathrm{sinc}^2(a\xi)\frac{\sin^2(\pi ML_0\xi)}{\sin^2(\pi L_0\xi)}$$

单缝衍射　　多光束干涉.

- 当 $\pi\xi L_0=M\pi$, 干涉因子分子分母均为0, 由洛必达法则: 干涉因子取极大值 M^2. 由此:

  $\xi=0,\ \pm\frac{1}{L_0},\ \pm\frac{2}{L_0},\ \cdots$ 即以空间频率表衍射强度取极大值位置

- 当干涉因子中分子为0, 分母 $\neq0$, 有极小值.

- 某一主极大与相邻2极小值间空间频率宽度为 $\Delta\xi$, 称为等效主极大宽度

$$\pi M \Delta \xi L_0 = \pm \pi \implies \Delta \xi = \frac{2}{M L_0}.$$

- 对应单、双、多缝可见：狭缝数↗，条纹越细.

衍射光栅

1. 线光栅.

无穷多平行狭缝构成.

$$t(x_0) = \sum_{m=-\infty}^{\infty} rect\left(\frac{x_0 - mL_0}{a}\right).$$

$$= rect\left(\frac{x_0}{a}\right) * \frac{1}{L_0} comb\left(\frac{x_0}{L_0}\right) \quad (卷积形式).$$

$$U_i(x_i) = \frac{e^{ikd} e^{\frac{ik}{2d}x_i^2}}{i\lambda d} F\{t(x_0)\}.$$

$$= \frac{a e^{ikd} e^{\frac{ik}{2d}x_i^2}}{iL_0} sinc\left(\frac{a x_i}{\lambda d}\right) \sum_{m=-\infty}^{\infty} \delta\left(x_i - \frac{m\lambda d}{L_0}\right).$$

衍射光强

$$I_i(x_i) = |U_i(x_i)|^2 = \left(\frac{a}{L_0}\right)^2 sinc^2\left(\frac{a x_i}{\lambda d}\right) \sum_{m=-\infty}^{\infty} \delta\left(x_i - \frac{m\lambda d}{L_0}\right).$$

由上可见，衍射±1级相对显著. 定义衍射效率 S

$$S = I_1 / I.$$

对无限多单狭缝光，其透射率在一半方波周期内为0，其余为1. 作 Fourier 展开：

$$t(x_0) = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi\xi_0 x_0) + \cdots = \frac{1}{2} + \frac{1}{\pi}\left(e^{i2\pi\xi_0 x_0} + e^{-i2\pi\xi_0 x_0}\right) + \cdots$$

其一级衍射效率：

$$S = \frac{1}{\pi^2} = 10.1\%$$

考虑实际光栅，屏有限大。若光栅为边长为L正方形.

$$t(x_0,y_0)=\left[\mathrm{rect}\left(\frac{x_0}{a}\right)*\frac{1}{L_0}\mathrm{comb}\left(\frac{x_0}{L_0}\right)\right]\mathrm{rect}\left(\frac{x_0}{L_0},\frac{y_0}{L_0}\right)$$

$$U_i(x_i,y_i)=\frac{e^{ikd}}{i\lambda d}e^{\frac{ik}{2d}(x_i^2+y_i^2)}F\{t(x_0,y_0)\}$$

$$=\frac{e^{ikd}e^{\frac{ik}{2d}(x_i^2+y_i^2)}}{i\lambda d}\frac{aL^2}{L_0}\sum_{m=-\infty}^{\infty}a\,\mathrm{sinc}\left(\frac{am}{L_0}\right)\mathrm{sinc}\left[L\left(\frac{x_i}{\lambda d}-\frac{m}{L_0}\right)\right]\mathrm{sinc}\left(\frac{Ly_i}{\lambda d}\right)$$

若观察平面上谱线间间隔$\frac{\lambda d}{L}$足够大，以至可忽略各衍射项衍射间重叠，